

ReactJS Hands-On Lab — Solution

Creating a React Application ("myfirstreact") and Displaying a Welcome Heading

Objectives

- Define SPA (Single-Page Application) and its benefits
- Define React and identify how it works
- Identify the differences between SPA and MPA
- Explain the pros and cons of a Single-Page Application
- Explain the basics of React
- Define the Virtual DOM
- Explain the features of React

In this hands-on lab, you will learn how to:

- Set up a React development environment
- Use create-react-app to scaffold a new project

Prerequisites

- Node.js
- NPM (installed automatically with Node.js)
- Visual Studio Code

Task

Create a new React application named "myfirstreact" and run it so that the browser displays "welcome to the first session of React" as the heading of the page.

Solution — Step by Step

Step 1: Install Node.js and NPM

Download and install Node.js (which bundles NPM) from the official site:

<https://nodejs.org/en/download/>

Verify the installation by running the following commands in a terminal / command prompt:

```
node -v
npm -v
```

Step 2: Install create-react-app globally

Run the following command in the command prompt to install the create-react-app tool:

```
npm install -g create-react-app
```

Step 3: Create the React application

Create a new React application named "myfirstreact" by running:

```
npx create-react-app myfirstreact
```

(create-react-app will download all the required packages and scaffold a new project folder named "myfirstreact".)

Step 4: Navigate into the project folder

Once the app has been created, move into the project folder:

```
cd myfirstreact
```

Step 5: Open the project in Visual Studio Code

Open the "myfirstreact" folder in Visual Studio Code. You can do this from within the folder by running:

```
code .
```

Step 6: Open App.js

In the Explorer panel of Visual Studio Code, navigate to the src folder and open the file App.js.

Step 7 & 8: Replace the contents of App.js

Delete the existing content of App.js and replace it with the following code:

```
function App() {  
  return (  
    <div className="App">  
      <h1>welcome to the first session of React</h1>  
    </div>  
  );  
}  
  
export default App;
```

Step 9: Run the application

Start the development server by running the following command inside the myfirstreact folder:

```
npm start
```

This compiles the application and automatically launches it in your default browser. If it doesn't open automatically, proceed to the next step.

Step 10: View the application in the browser

Open a new browser window and navigate to:

```
http://localhost:3000
```

The page will display "welcome to the first session of React" as a heading (rendered from the <h1> tag), confirming that the React application has been successfully created and is running.

Result

A new React application named "myfirstreact" was successfully created using create-react-app, the App.js component was modified to render an `<h1>` heading with the text "welcome to the first session of React", and the application was verified by running it locally and viewing it in the browser at <http://localhost:3000>.